INTRODUCTION

Overview

What makes it unique is the complexity and uncertainty involved when interacting with the natural geological environment.

Rock masses are complex systems!

Often, field data (e.g. geology, geological structure, rock mass properties, groundwater, etc.) is limited to surface observations and/or limited by inaccessibility, and can never be known completely.

Nature of Rock

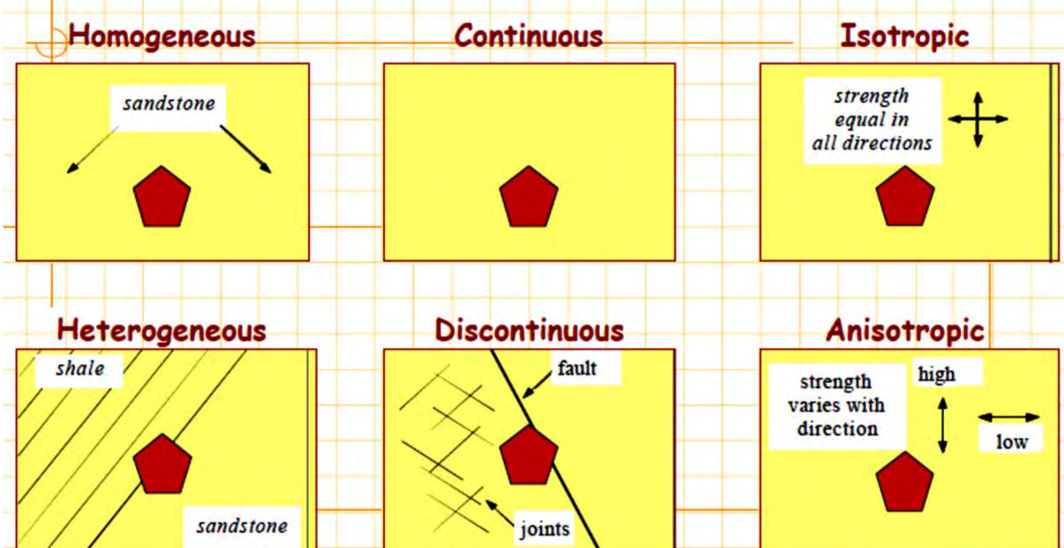
A common assumption when dealing with the mechanical behaviour of solids is that they are:

- homogeneous
- continuous
- isotropic

However, rocks are much more complex than this and their physical and mechanical properties vary according to scale. As a solid material, rock is often:

- heterogeneous
- discontinuous
- anisotropic

Nature of Rock



CHILE

CONTINUOUS

HOMOGENEOUS

ISOTROPIC

LINEAR **E**LASTIC

DIANE

DISCONTINUOUS

➡ pores/microfractures - vugs, joints - faults, caverns

INHOMOGENEOUS

➡ mineralogy-layering-facies

ANISOTROPIC

➡ fabric - mineral alignment-discontinuity sets

NON **E**LASTIC

Rock as an Engineering Material

One of the most important, and frequently neglected, aspects of rock mechanics and rock engineering is that we are utilizing an existing material which is usually highly variable.

intact

'layered' intact

highly fractured

Rock as an Engineering Material

Rock as an engineering material will be used either:

… as a building material so the structure will be made *of* rock
… or a structure will be built *on* the rock
… or a structure will be built *in* the rock

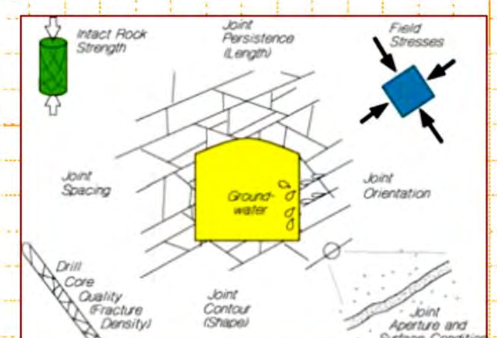
In the context of the mechanics, we must establish:

… the properties of the material
… the pre-existing stress state in the ground (which will be disturbed by the structure)
… and how these factors relate to the engineering objective

Influence of Geological Factors

In the context of the mechanics problem, we should consider the material and the forces involved. As such, five primary geological factors can be viewed as influencing a rock mass.

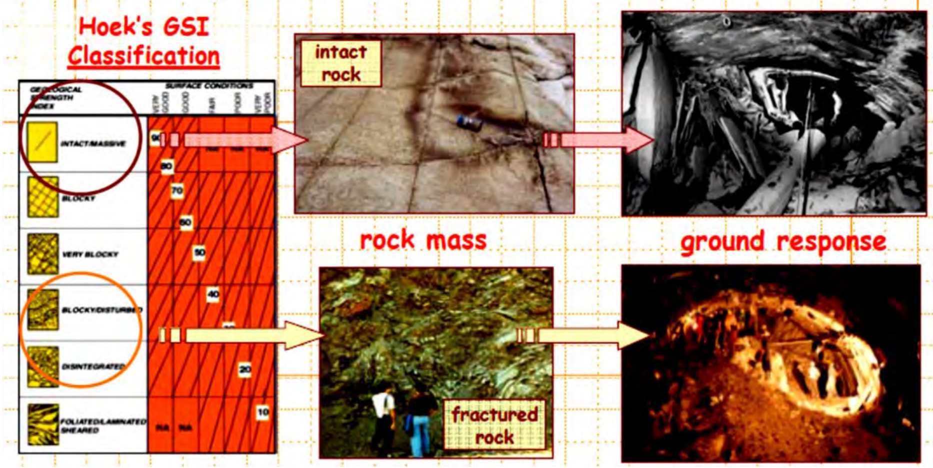
- We have the intact rock which is itself divided by discontinuities to form the rock mass structure.
- We find then the rock is already subjected to an *in situ* stress.
- Superimposed on this are the influence of pore fluid/water flow and time.



With all these factors, the geological history has played its part, altering the rock and the applied forces.

Rock as an Engineering Material

The key factor that distinguishes rock engineering from other engineering-based disciplines is the application of mechanics on a large scale to a pre-stressed, naturally occurring material.

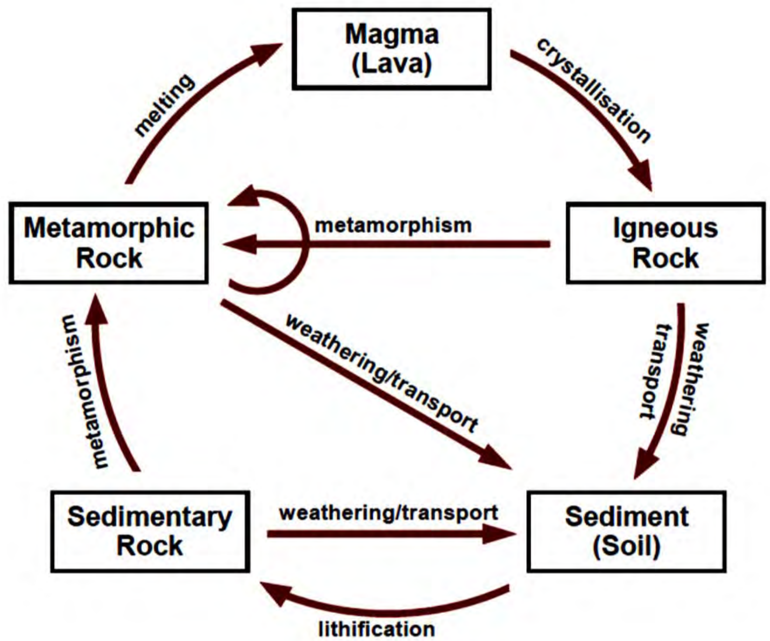


Rock Formation

Origin of Rock

Rock is a natural solid substance composed of minerals.

Rocks are formed by three origins: *igneous* rocks from magma, *sedimentary* rock from sediments lithification and *metamorphic* rocks through metamorphism, as illustrated by the rock cycle.



Rock Formation

Minerals

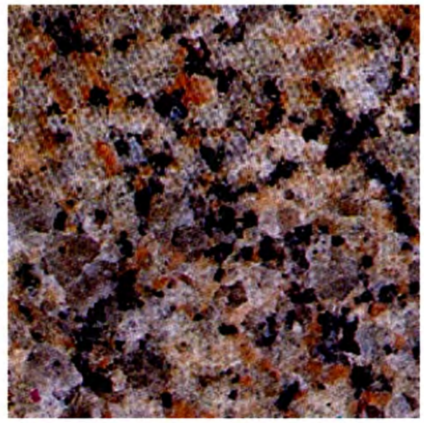
Rocks are composed of minerals, primarily silicates. Important rock-forming silicates are feldspars, quartz, olivines, pyroxenes, amphiboles, garnets, and micas.

Minerals have different properties, crystal structure, hardness and cleavage influence rock properties.

In rock, mineral crystals are often massive, granular or compact, and only microscopically visible.

Rock Formation

Well developed quartz crystal



Quartz in granite

Rock Formation

Igneous Rocks

Igneous rocks are formed when molten rock (magma) cools and solidifies, with or without crystallization.

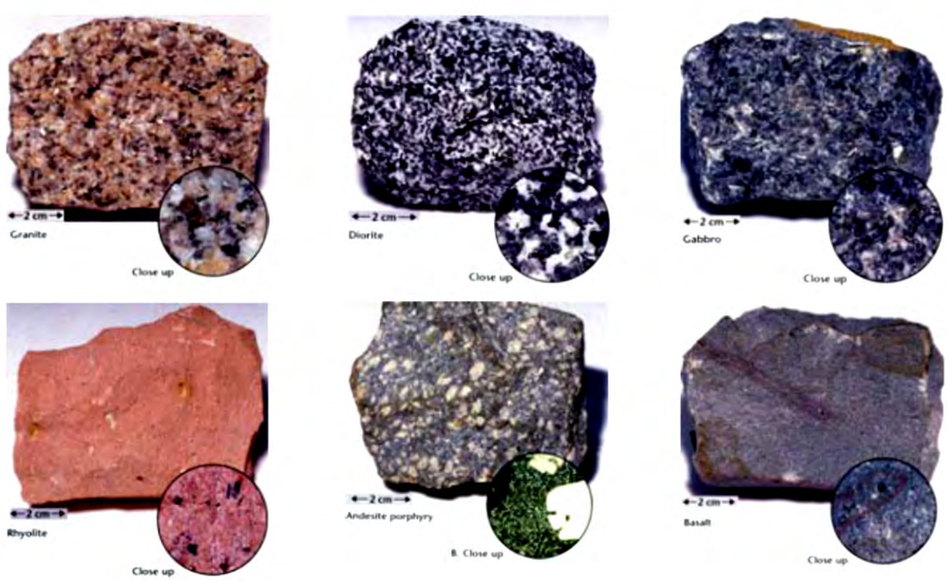
They can be formed (i) below the surface as intrusive (plutonic) rocks, or (ii) on the surface as extrusive (volcanic) rocks. Intrusive is generally coarse grained and extrusive fine grained.

They can also have different mineral contents.

Rock Formation

	Granitic (acid) (felsic)	Andesitic (intermediate)	Basaltic (basic) (mafic)	Ultramafic (ultrabasic)
Intrusive (coarse grain)	Granite	Diorite	Gabbro	Peridotite
Extrusive (fine grain)	Rhyolite	Andesite	Basalt	None
Silica Content	>65% Silica	50-65% Silica	40-50% Silica	<40% Silica
Main Mineral Composition	Quartz Orthoclase N-Plagioclase	Amphibole Plagioclase Biotite	Ca-Plagioclase Pyroxene	Olivine Pyroxene
Minor Mineral Composition	Muscovite Biotite Amphibole	Pyroxene	Olivine Amphibole	Ca-Plagioclase
Colour	Light	→	→	Dark

Rock Formation



Rock Formation

Sedimentary Rocks

Sedimentary rock is formed in three main ways:
(i) deposition of the weathered remains of other rocks (known as 'clastic' sedimentary rocks);
(ii) deposition of the results of biogenic activity; and
(iii) precipitation from solution.

Clastic sedimentary rocks are commonly classified by grain size.

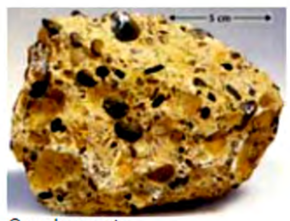
Rock Formation

Particle size	Comments	Rock name
> 2 mm	Rounded rock fragment	Conglomerate
	Angular rock fragment	Breccia
1/16 - 2 mm	Quartz with other minerals	Sandstone
< 1/16 mm	Split into thin layers	Shale
	Break into clumps or blocks	Mudstone

Rock Formation

Shale

Sandstone



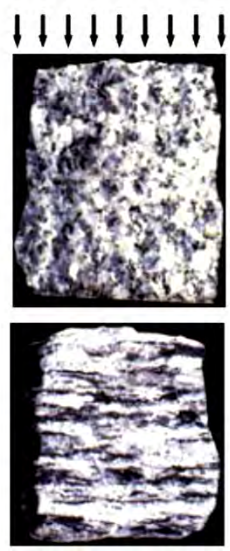
Conglomerate

Rock salt

Rock Formation

Metamorphic Rocks

Metamorphic rock is a new rock transformed from an existing rock, through metamorphism – change due to heat and pressure.



Metamorphic rocks can have foliated and non-foliated textures. Foliation is due to the re-orientation of mica minerals, creating a plane of cleavage or visible mineral alignment feature.

Rock Formation

Rock	Texture	Metamorphic grade	Original parent rock
Slate	Foliated	Low grade	Shale (clay minerals)
Phyllite	Foliated	Low to intermediate grade	Shale
Mica schist	Foliated	Low to intermediate grade	Shale
Chlorite schist	Foliated	Low grade	Basalt
Gneiss	Foliated	High grade	Granite, shale, andesite
Marble	Non-foliated	Low to high grade	Limestone, dolomite
Quartzite	Non-foliated	Intermediate to high grade	Quartz sandstone

Rock Formation

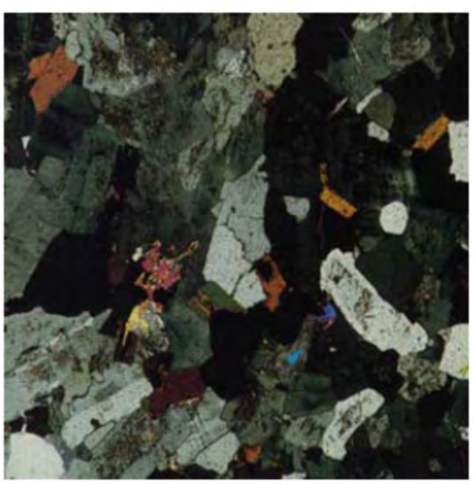
Rock Formation

Rock Textures

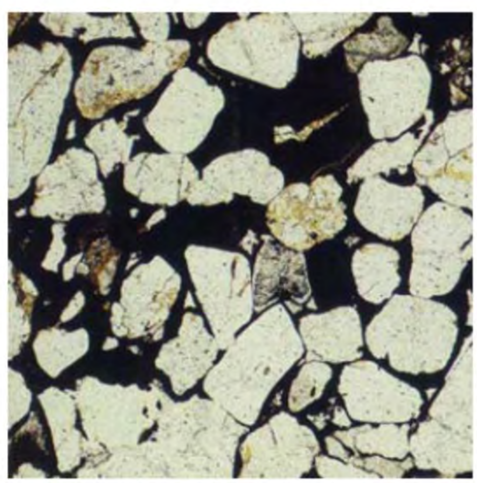
Sedimentary, igneous and metamorphic rocks have different textures due to their different origin. The two main texture forms are clastic and interlocking.

Rock material strength is a structural strength of the composition of the minerals. It is governed by
(i) the strength of the minerals, and
(ii) the structural bonding (integration) of the minerals.

Rock Formation



Interlocking structure of a granite



Clastic structure of a sandstone

Rock Formation

Rock Textures

The interlocking microstructures of igneous and metamorphic rocks lead to generally high strength of rock material, while the clastic microstructures of sedimentary rocks often lead to low rock material strength, particularly when cementation is weak.

Any existing weakness in a rock material matrix (microcracks, pores, and weak grains and cementation) also weakens the rock material.

Rock Discontinuities

Rock Joints

Joints are the most common rock discontinuity. They are normally in parallel sets.

They are generally considered as part of the rock mass. The spacing of joints is usually in the order of a few to a few ten centimetres. For engineering, joints are constant features of the rock mass.

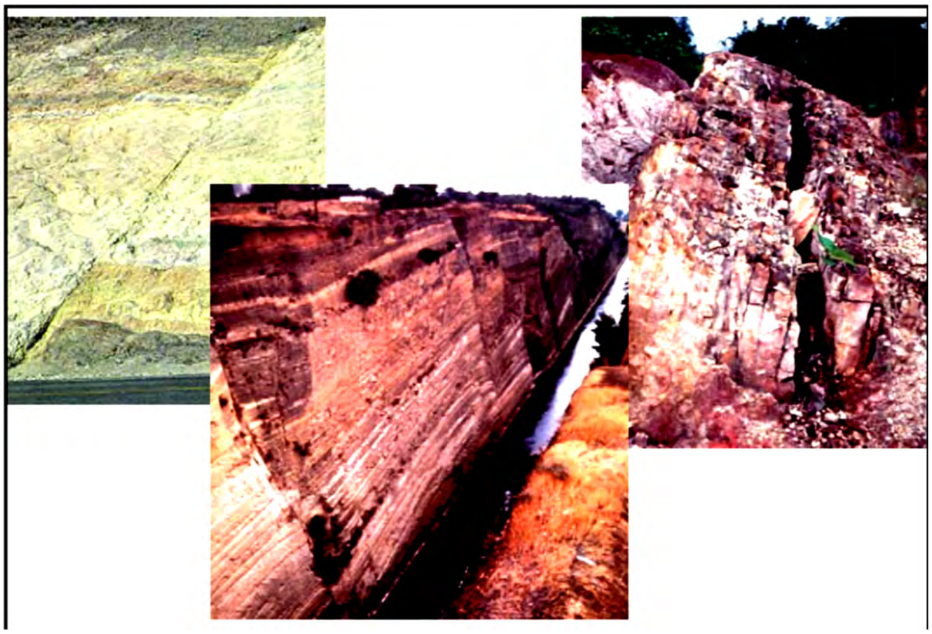
Rock Discontinuities

Rock Discontinuities

Faults

Faults are planar rock fractures which show evidence of relative movement. Faults have different scale and the largest faults are at tectonic plate boundaries. Faults usually do not consist of a single, clean fracture, they often form fault zones.

Large scale fault, fault zone and shear zone, are large and localised feature. They are often dealt separately from the rock mass.

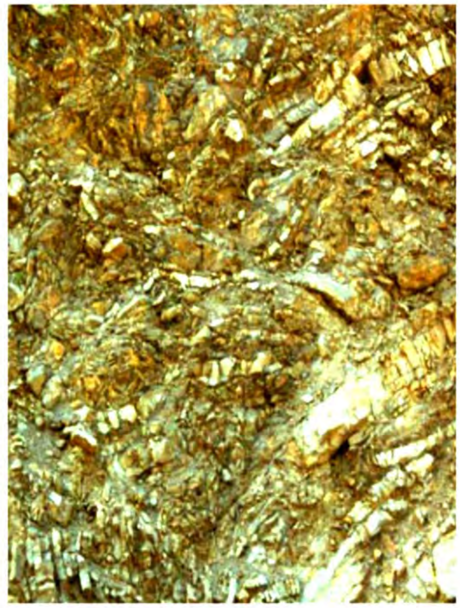


Rock Discontinuities

Folds

Fold is the bended originally flat and planar rock strata, as a result of tectonic force or movement.

Folds are usually not considered as part of the rock mass. They are often associated with high degree of fracturing and relatively weak and soft rocks.

Rock Discontinuities

Bedding Planes

Bedding plane is the interface between sedimentary rock layers.

Bedding planes are isolated geological features to engineering activities. It mainly creates an interface of two rock materials. However, some bedding planes could also become potential weathered zones and groundwater pockets.

Rock Material and Rock Mass

Engineering Scale of Rock

For civil engineering works, e.g., foundations, slopes and tunnels, the scale of projects is usually a few tens to a few hundreds metres.

Rock in an engineering scale is generally a mass of rock at the site. This mass of rock, often termed as rock mass, is the whole body of the rock in situ, consists of intact rock blocks and all types of discontinuities (joints, faults etc).

Rock Material and Rock Mass

Composition of Rock Mass

A rock mass contains (i) rock material, in the form of intact rock blocks of various sizes, and (ii) rock discontinuities that cuts through the rock, in the forms of fractures, joints, faults, bedding planes, and dykes.

Rock mass = Rock materials + Rock discontinuities

Rock Material and Rock Mass

Roles of Rock Joints in Rock Mass Behaviour

- Cuts rock into slabs, blocks and wedges, to be free to fall and move;
- Acts as weak planes for sliding and moving;
- Provides water flow channel and creates flow networks;
- Gives large deformation;
- Alters stress distribution and orientation;

Rock mass behaviour is largely governed by joints.

Inhomogeneity and Anisotropy

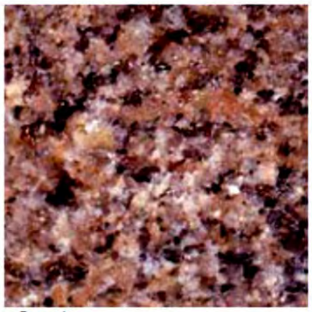
Inhomogeneity of Rock Material

Inhomogeneity represents property varying with locations. Many construction materials have varying degrees of inhomogeneity. Rock is formed by nature and exhibits great inhomogeneity, due to:

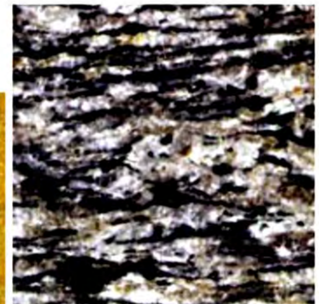
(i) different minerals in a rock,
(ii) different bounding between minerals,
(iii) existence of pores,
(iv) existence of microcracks.

Inhomogeneity and Anisotropy

Texture of some common rocks


Granite


Sandstone


Gneiss

Inhomogeneity and Anisotropy

Inhomogeneity of Rock Material

Inhomogeneity is the cause of fracture initiation leading to the failure of a rock material. If some elements in the rock material matrix are very weak, they will start to fail early and usually lead to low overall strength of the rock material.

Inhomogeneity and Anisotropy

Inhomogeneity of Rock Mass

Inhomogeneity of a rock mass is primarily due to the existence of the various discontinuities.

Rock masses are also inhomogeneous due to the mix of rock types, interbedding and intrusion.

Inhomogeneity and Anisotropy

Anisotropy

Anisotropy is defined as properties are different in different direction. It occurs in both rock materials and rock mass.

Rock with obvious anisotropy is slate. Metamorphic phyllite and schist and sedimentary shale also exhibit anisotropy.

Inhomogeneity and Anisotropy

Anisotropy

Rock mass anisotropy is controlled by
(i) joint set, and
(ii) sedimentary layer.

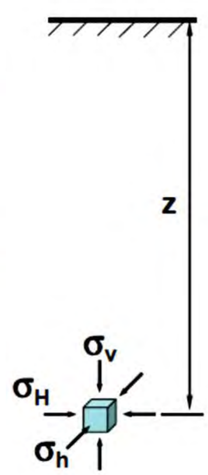
In Situ Stresses

Vertical Stress and Overburden

At depth, vertical stress in rock is the overburden stress generated by weight of the overlying material.

The average specific gravity of rocks is about 2.7. The vertical stress at depth can be estimated as

σ_v (MPa) ≈ 0.027 z (m)



In Situ Stresses

Horizontal Stress and Tectonic Stress

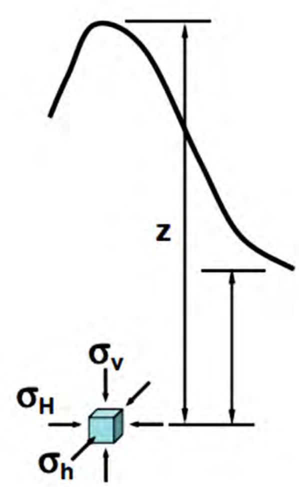
Horizontal stresses in rock are primarily tectonic stress.

Horizontal stresses in rocks are generally higher than vertical stress. The maximum horizontal stress is usually in the same directions as tectonic convergence movement. Tectonic stress has huge variations in magnitude, and can be exceptionally large.

In Situ Stresses

In situ stress field can also be altered by geological factors and processes:

- **Surface topography**
- **Erosion**
- **Intrusion**
- **Fault and faulting**



In Situ Stresses

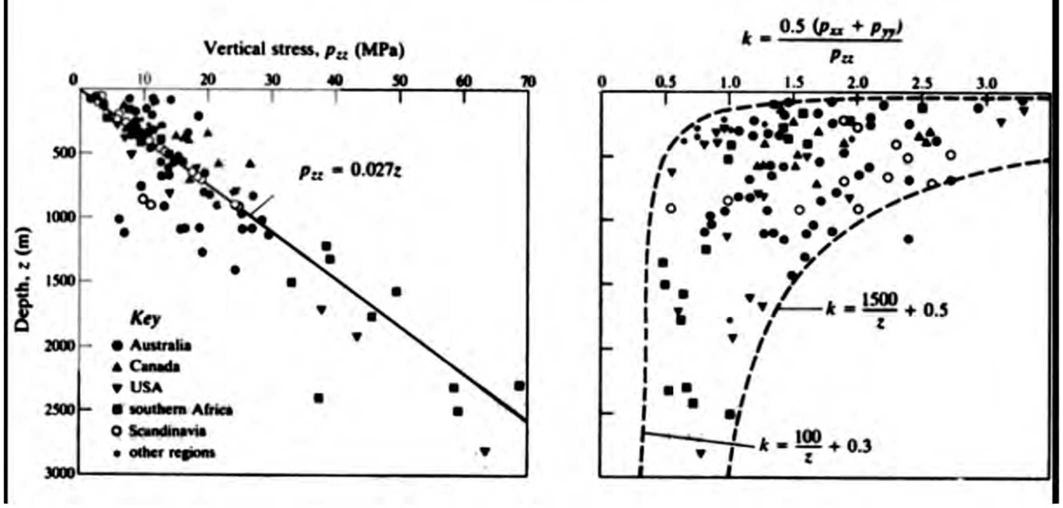
In situ Stress Measurements

In situ stress measurements show that vertical stress is about 0.027z, the overburden pressure.

Ratio of average horizontal stresses $(\sigma_H+\sigma_h)/2$ to vertical stress is between 0.5 to 3.0, mostly bounded between (100/z +0.3) and (1500/z +0.5).

At common depth for civil engineering (<1000 m), the variation of horizontal stress is wide.

In Situ Stresses



In Situ Stresses

In rock, one horizontal stress is usually the major principal stress, while the vertical stress or the other horizontal stress represents the minor principal stress, i.e.,

$\sigma_H > \sigma_h > \sigma_v$ or $\sigma_H > \sigma_v > \sigma_h$

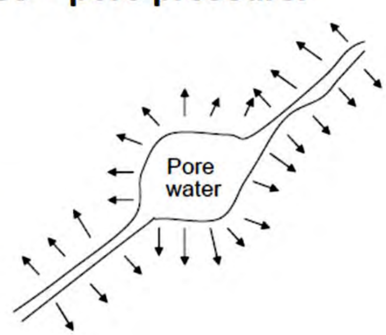
Vertical stress can be estimated from overburden. Horizontal stresses should not be estimated. If horizontal stress directions and magnitudes are needed, in situ stress measurements must be conducted.

In Situ Stresses

Effective Stress

In porous material, e.g., sandstone, effective stress may be computed as total stress – pore pressure.

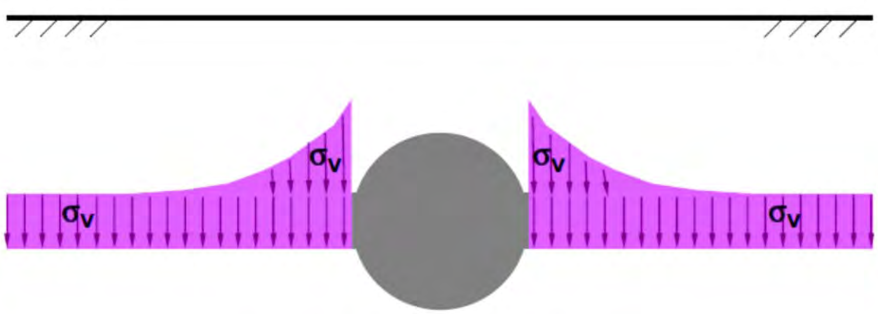
In fractured rock mass, distribution of water is no longer even and stress field is no longer uniform. Hence, the effective stress principle is no longer applicable.



In Situ Stresses

Re-distribution of Stress

Rock engineering is an activity disturbing the original stress field which is already in equilibrium. Rock mechanics deals with stress re-distribution and redistributed stresses, and the short term response of rock during stress re-distribution and long term behaviour in the redistributed stress field.



Ground Water

Flow in Rock Material

Most of the igneous and metamorphic rocks are very dense with interlocked texture. The rocks therefore have extremely low permeability and porosity. Some clastic sedimentary rocks, typically sandstones, can be porous and permeable. Weathered rocks can also be porous and permeable.

Ground Water

Flow in Fracture Network

Rock masses are fractured. Fractures provide flow paths and flow is governed by the apertures.

Flow in a fractured rock mass is also controlled by the connectivity of fracture system or network. Although a rock mass can be seen as highly fractured, only a limited percentage of fractured are interconnected and conduct flow. At site it is common to see only a few fractured has water flow while others are dry.

Ground Water

Effects of Groundwater and Pressure

Groundwater is important to rock mechanics:

(i) Water pressure contributes to the stress field;
(ii) Water changes rock parameters, e.g., friction;
(iii) When water is present, it increases the complexity of rock engineering, e.g., more difficult to tunnel with water inflow and high water pressure.

Special Rocks

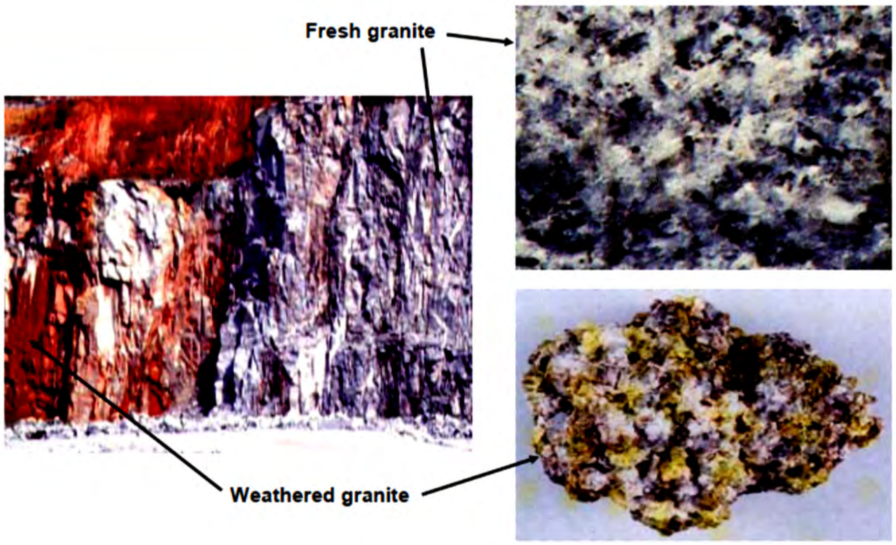
Weathering and Weathered Rocks

All rocks disintegrate slowly as a result of:

(i) Mechanical weathering, breakdown of rock into particles without changing chemical composition of the minerals in the rock.

(ii) Chemical weathering, breakdown of rock by chemical reaction, primarily by water and air.

Special Rocks



Special Rocks

Weathered Rock

Weathering is progressive, between fresh rock and completed material (soil), rocks can be slightly, moderately and highly weathered. Those weathered rocks are still intact and have structure and texture as rock. However, due to weathering, their properties have been affected and altered.

Weathering causes significant reduction of rock material strength.

Special Rocks

Soft Rocks and Hard Soils

Sedimentary rocks are formed by sediments (soils) through long processes of compaction and cementation. The process could be stopped before the sediments are being completed solidified. The materials then could be highly consolidated but not fully solidified. Typically, those materials have low strength and high deformability, and when placed in water, they often can be dissolved. When dry , they behave as weak rock and when in water, it collapses.

Special Rocks

Special Rocks

Swelling Rock

Some rocks have the characteristics of swelling, that is when the rock is exposed with water (directly in contact with water or in air), it expanse. This is primarily due the swelling behaviour of the minerals of the rock, typically the montmorillonite clay mineral. Rock and soil containing considerable amount of montmorillonite minerals will exhibit swelling and shrinkage characteristics.

Special Rocks

Crushed Rock

Characteristics of highly fractured and crushed rocks are quite different from the massive rock mass. They behave as granular and block materials, depending on the geometry and friction. When such materials are encountered in engineering, they need to be addresses separately.

Special Rocks